

The critical fractional bandlimiting projection

A full answer to the projection question in arXiv:2002.12015

Automated mathematics run `fa_banach_001`, lane 4

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Status. Candidate full solution of the critical bandlimiting-projection question, under the canonical homogeneous quotient definition or, equivalently, the critical BMO realization recorded by the source. The separate problem of identifying a preferred critical *interpolation* normalization remains open. Human expert review is recommended before dissemination.

1 Source question

Monguzzi, Peloso and Salvatori introduce fractional Paley–Wiener and Bernstein spaces and prove boundedness of the Hilbert bandlimiting projection away from the critical indices. In Section 8 they ask whether the projection is bounded when

$$s - \frac{1}{p} \in \mathbb{N}_0,$$

and suggest defining the critical space through the homogeneous Sobolev class modulo polynomials [1]. The displayed question is reproduced below from page 23 of the arXiv PDF.

8. FINAL REMARKS AND OPEN QUESTIONS

We believe that the fractional spaces we introduced are worth investigating and, as we mentioned, they arise naturally in a several variable setting ([MPS20b]).

We mention a few questions that remain open. First of all, it is certainly of interest to consider the cases $s - \frac{1}{p} \in \mathbb{N}_0$. As we pointed out already, these cases correspond to the critical cases in the Sobolev embedding theorem. As shown by Bourdaud, when $s - \frac{1}{p} \notin \mathbb{N}_0$, the realization spaces $E^{s,p}$ of $\dot{W}^{s,p}$ are the unique realization spaces whose norms are homogeneous with respect the natural dilations. On the other hand, when $s - \frac{1}{p} \in \mathbb{N}_0$ there exists no realization space of $\dot{W}^{s,p}$ whose norm is homogeneous. In these case, it would be natural to define the realization space as the interpolating space between two spaces with $s - \frac{1}{p} \notin \mathbb{N}_0$. Thus, a natural definition may be

$$\mathcal{B}_a^{s,p} = \{f \in \mathcal{E}_a : [f_0]_m \in \dot{W}^{s,p} \text{ and if } m \geq 1/p, P_{f_0;m;0} = 0\},$$

where $[f_0]_m$ denotes the equivalence class of f_0 in $\mathcal{S}'/\mathcal{P}_m$. In any event, these spaces remain to be investigated. Naturally, another question that remains open is the boundedness of the orthogonal projection $\mathbf{P} : E^{s,p} \rightarrow \mathcal{B}_a^{s,p}$ in the cases $s - \frac{1}{p} \in \mathbb{N}_0$. Such boundedness would allow one to explicitly describe the dual space of $\mathcal{B}_a^{s,p}$, for the whole scale $s > 0$ and $p \in (1, \infty)$.

The Paley–Wiener space is a very special instance of a de Branges spaces. These spaces where

The source also stresses a genuine issue: at a critical index there is no dilation-homogeneous concrete realization. There is nevertheless a canonical homogeneous *quotient*, and the BMO/Taylor conditions of [2] select a concrete representative. The theorem below treats both descriptions and keeps their normalizations separate.

2 Definitions and statement

Fix $a, s > 0$ and $1 < p < \infty$, and suppose

$$m = s - \frac{1}{p} \in \mathbb{N}_0.$$

Let $\dot{W}^{s,p}$ be the completion of $\mathcal{S}(\mathbb{R})$ for the norm $\|\Delta^{s/2}f\|_p$, viewed in $\mathcal{S}'/\mathcal{P}_m$. Let $E^{s,p}$ be the critical realization from [2, Theorem 3(iii)]: for $m = 0$ it is a fixed-mean representative in BMO, and for $m \geq 1$ it is the corresponding Taylor/Sobolev–BMO representative. Write

$$U_{s,p} : E^{s,p} \longrightarrow L^p(\mathbb{R}), \quad U_{s,p}f = \Delta^{s/2}f.$$

The realization theorem and the definition of $\dot{W}^{s,p}$ make $U_{s,p}$ a surjective isometry; a short proof is included below.

Let Q_a denote the interval Fourier projection

$$Q_ag = \mathcal{F}^{-1}(\mathbf{1}_{[-a,a]}\hat{g}).$$

It is interpreted as the standard L^p Fourier multiplier, not as an undefined product of an arbitrary distribution by a discontinuous function. In one dimension Q_a is a difference of modulated Riesz projections, hence is bounded for $1 < p < \infty$.

Define the critical bandlimited realization

$$\tilde{\mathcal{B}}_{a,\text{crit}}^{s,p} = \{f \in E^{s,p} : \text{supp } \hat{f} \subseteq [-a, a]\}.$$

Theorem 1 (Critical projection theorem). *For every $a, s > 0$ and $1 < p < \infty$ with $s - 1/p \in \mathbb{N}_0$:*

1. *Every element of $\tilde{\mathcal{B}}_{a,\text{crit}}^{s,p}$ is the restriction, in the sense of tempered distributions, of an entire function of exponential type at most a .*
2. *$U_{s,p}$ restricts to a surjective isometry*

$$U_{s,p} : \tilde{\mathcal{B}}_{a,\text{crit}}^{s,p} \longrightarrow \mathcal{B}_a^p,$$

where $\mathcal{B}_a^p$ is the classical L^p Bernstein space.

3. *The operator*

$$\Pi_{a,s,p} = U_{s,p}^{-1}Q_aU_{s,p}$$

is a bounded projection of $E^{s,p}$ onto $\tilde{\mathcal{B}}_{a,\text{crit}}^{s,p}$, and

$$\|\Pi_{a,s,p}\|_{E^{s,p} \rightarrow E^{s,p}} = \|Q_a\|_{L^p \rightarrow L^p}.$$

4. *If $p = 2$, then $\Pi_{a,s,2}$ is the Hilbert-space orthogonal projection. On the common dense domain used in [1], the operator for general p agrees with the source projection modulo $\mathcal{P}_m$; it agrees exactly after either the critical realization normalization or the Taylor normalization is imposed.*

Consequently the boundedness question in Section 8 of [1] has an affirmative answer for the canonical critical quotient and for its critical BMO realization.

3 Proof intuition

The fractional Laplacian is the correct coordinate map. On the homogeneous Sobolev quotient it is an isometric isomorphism onto L^p . Bandlimiting after this change of coordinates is simply the familiar interval multiplier Q_a . Conjugating Q_a therefore gives a bounded idempotent immediately.

Two checks turn this operator-theoretic observation into an answer about entire functions. First, if $\Delta^{s/2}f$ is bandlimited, then $\widehat{f}$ vanishes outside the same interval: away from frequency zero one may divide test functions by $|\xi|^s$. The possible ambiguity at zero is only a polynomial, whose Fourier transform is supported at zero. Second, a tempered distribution with compactly supported Fourier transform is the restriction of an entire function of the corresponding exponential type by Paley–Wiener–Schwartz. Thus no borderline Sobolev embedding estimate is needed.

The critical BMO normalization and a Taylor normalization generally select different polynomial representatives. Subtracting the Taylor polynomial only *after* bandlimiting gives an isometric identification between them and avoids imposing two incompatible normalizations at once.

4 Proof

4.1 The fractional Laplacian is onto

Let $\mathcal{S}_\infty$ be the Schwartz functions all of whose moments vanish. It is dense in L^p for $1 < p < \infty$ [2]. For $g \in \mathcal{S}_\infty$, the homogeneous Riesz potential

$$I_s g = \mathcal{F}^{-1}(|\xi|^{-s} \widehat{g})$$

again lies in $\mathcal{S}_\infty$, and $\Delta^{s/2} I_s g = g$. Given $h \in L^p$, choose $g_n \in \mathcal{S}_\infty$ with $g_n \rightarrow h$ in L^p . Then $I_s g_n$ is Cauchy in the defining norm of $\dot{W}^{s,p}$ and determines a class $[f]$ satisfying

$$\Delta^{s/2}[f] = h, \quad \|[f]\|_{\dot{W}^{s,p}} = \|h\|_p.$$

The realization map of [2] selects the unique representative $f \in E^{s,p}$ without changing the norm. Hence $U_{s,p}$ is a surjective isometry.

4.2 The support lemma

Lemma 2. *For $f \in E^{s,p}$,*

$$\text{supp } \widehat{U_{s,p} f} \subseteq [-a, a] \iff \text{supp } \widehat{f} \subseteq [-a, a].$$

Proof. The distributional fractional-Laplacian identity can be tested without any ambiguity on functions whose Fourier support avoids zero. Suppose first that $g = U_{s,p} f$ and $\text{supp } \widehat{g} \subseteq [-a, a]$. If $\psi \in C_c^\infty(\mathbb{R} \setminus [-a, a])$, then $\theta(\xi) = |\xi|^{-s} \psi(\xi)$ is again a smooth compactly supported test function. Therefore

$$\langle \widehat{f}, \psi \rangle = \langle \widehat{f}, |\xi|^s \theta \rangle = \langle \widehat{g}, \theta \rangle = 0.$$

This proves $\text{supp } \widehat{f} \subseteq [-a, a]$.

Conversely, take a test function ψ supported outside $[-a, a]$. Multiplication or division by $|\xi|^s$ is smooth on its support, so the same identity, in the opposite direction, gives $\langle \widehat{g}, \psi \rangle = 0$. Notice that changing f by a polynomial adds only derivatives of δ_0 to $\widehat{f}$ and therefore cannot affect either conclusion. $\square$

If $f \in \tilde{\mathcal{B}}_{a,\text{crit}}^{s,p}$, Lemma 2 says that $U_{s,p}f$ is an L^p function with Fourier support in $[-a, a]$. The classical Bernstein-space characterization makes it an element of $\mathcal{B}_a^p$. Conversely, if $g \in \mathcal{B}_a^p$, let $f = U_{s,p}^{-1}g$. Lemma 2 gives $\text{supp } \hat{f} \subseteq [-a, a]$. Since $f \in \mathcal{S}'$ and its Fourier transform is compactly supported, the Paley–Wiener–Schwartz theorem gives an entire function F satisfying

$$|F(z)| \leq C(1 + |z|)^N e^{a|\text{Im } z|}$$

for some C, N , whose restriction equals f as a distribution. The polynomial factor is absorbed by $e^{\varepsilon|z|}$ for every $\varepsilon > 0$, so this is exponential type at most a in the convention of [1]. This proves parts 1 and 2 of Theorem 1.

4.3 Conjugating the interval multiplier

The interval characteristic function is an L^p multiplier for $1 < p < \infty$: Q_a is a difference of two modulated half-line Riesz projections. It is an idempotent whose range is precisely the L^p functions with distributional Fourier support in $[-a, a]$. Since $U_{s,p}$ is a surjective isometry,

$$\Pi_{a,s,p} = U_{s,p}^{-1}Q_a U_{s,p}$$

is an idempotent, has range $U_{s,p}^{-1}(\mathcal{B}_a^p) = \tilde{\mathcal{B}}_{a,\text{crit}}^{s,p}$, and satisfies

$$\|\Pi_{a,s,p}\| = \|Q_a\|_{p \rightarrow p}.$$

This proves part 3.

When $p = 2$, $U_{s,2}$ is unitary and Q_a is the orthogonal projection in L^2 ; unitary conjugation proves the first assertion of part 4. For a Schwartz input f , away from zero the Fourier transform of $\Pi_{a,s,p}f$ is $\mathbf{1}_{[-a,a]}\hat{f}$. The source projection has the same Fourier transform there. Their difference is supported at zero and is therefore a polynomial; in the homogeneous space the two operators are identical. The uniqueness of the chosen realization representative (or of a chosen Taylor representative) then makes the equality exact. This completes the proof of Theorem 1.

5 The quotient and Taylor-normalized formulations

Define the homogeneous critical Bernstein space by

$$\dot{\mathcal{B}}_{a,\text{crit}}^{s,p} = \{[F_0]_m \in \dot{W}^{s,p} : F \text{ is entire of type } \leq a\}, \quad \|[F_0]_m\| = \|\Delta^{s/2}F_0\|_p.$$

The preceding proof shows directly that it is isometric to both $\tilde{\mathcal{B}}_{a,\text{crit}}^{s,p}$ and $\mathcal{B}_a^p$.

If an actual entire representative is desired, take the BMO-normalized bandlimited representative $f \in E^{s,p}$, let F be its entire extension, and set

$$J_m F = F - P_{F;m;0}.$$

Then $J_m F$ has the same type, represents the same class, has the same norm, and satisfies $P_{J_m F;m;0} = 0$. Thus J_m is a linear isometric bijection from the critical BMO-normalized bandlimited realization to the Taylor-normalized model. In this model the bounded map asked for in the source is $J_m \Pi_{a,s,p}$; it is the same projection on the homogeneous quotient. For $m = 0$, one either subtracts the value at zero or works modulo constants. This also resolves the minor normalization ambiguity in the source's displayed tentative definition.

Corollary 3 (Dual description). *With $p' = p/(p - 1)$, the dual of the critical bandlimited space is canonically*

$$(\tilde{\mathcal{B}}_{a,\text{crit}}^{s,p})^* \cong L^{p'}(\mathbb{R})/\ker Q_a^*,$$

under the pairing $\langle f, h \rangle = \int (U_{s,p} f) \bar{h}$. Since the multiplier is real and $Q_a^ = Q_a$ on $L^{p'}$, this quotient is topologically identified with the classical bandlimited subspace $\mathcal{B}_a^{p'}$ (with the projection-induced quotient norm).*

6 Verification, limitations and novelty

- The proof uses the critical realization theorem of [2], the density/invertibility of the fractional Laplacian on $\mathcal{S}_\infty$, the one-dimensional interval multiplier theorem, and Paley–Wiener–Schwartz. No unproved interpolation assertion is used.
- The fixed-mean BMO normalization and the Taylor normalization are never imposed simultaneously. They are related only after the output is known to be entire.
- The result covers exactly $1 < p < \infty$. The interval projection is not bounded on L^1 or L^∞ , so this argument correctly does not claim the endpoints.
- The theorem settles the bounded projection for the canonical quotient and the source’s recorded critical realization. It does not prove that a particular concrete normalization is the complex or real interpolation space between neighboring noncritical spaces.
- A bounded search on 11 August 2026 covered the run’s registry, solution, attempt and proof-gap indexes; the exact title and arXiv id; the phrases “ $s - 1/p$ Fractional Paley–Wiener”, “critical fractional Bernstein projection”, and close variants; the Springer article; and visible citing followups. The published article still labels the question open. No later paper answering the exact critical projection question was found. Novelty confidence is moderate, not definitive.

Primary reviewer focus. Check the support identity in Lemma 2, the use of the realization isomorphism at critical indices, and the exact representative convention when comparing with the source Hilbert projection.

References

- [1] A. Monguzzi, M. M. Peloso and M. Salvatori, *Fractional Paley–Wiener and Bernstein spaces*, Collect. Math. **72** (2021), 615–643; arXiv:2002.12015; doi:10.1007/s13348-020-00303-4.
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